When X = 8 the value of Y is estimated as

$$Y = 0.8(8)+2.6$$

$$Y = 9$$

Example 9.17

The two regression lines are $3X+2Y=26$ and $6X+3Y=31$. Find the correlation coefficient.

Solution:

Let the regression equation of Y on X be

$$3X+2Y = 26$$

$$2Y = -3X+26$$

$$Y = \frac{1}{2}(-3X+26)$$

$$Y = -1.5X+13$$

$$r\frac{\sigma_y}{\sigma_x} = -1.5$$

Implies by $x = r\frac{\sigma_y}{\sigma_x} = -1.5$

Let the regression equation of X on Y be

$$6X+3Y = 31$$

$$X = \frac{1}{6}(-3Y+31) = -0.5Y+5.17$$

$$r\frac{\sigma_x}{\sigma_y} = -0.5$$

Implies $b_{xy} = r\frac{\sigma_x}{\sigma_y} = -0.5$

$$r = \pm\sqrt{b_{xy}.b_{yx}}$$

$= -\sqrt{(-1.5).(-0.5)}$ (Since both the regression coefficient are negative r is negative)

$$\therefore\ r = -0.866$$

Example 9.18

In a laboratory experiment on correlation research study the equation of the two regression lines were found to be $2X-Y+1=0$ and $3X-2Y+7=0$. Find the means of X and Y. Also work out the values of the regression coefficient and correlation between the two variables X and Y.

Solution:

Solving the two regression equations we get mean values of X and Y.

$$2X-Y = -1 \quad \ldots (1)$$

$$3X-2Y = -7 \quad \ldots (2)$$

Solving equation (1) and equation (2) We get $X=5$ and $Y=11$.

Therefore the regression line passing through the means $\overline{X}=5$ and $\overline{Y}=11$.

The regression equation of Y on X is $3X-2Y=-7$

$$2Y = 3X+7$$

$$Y = \frac{1}{2}(3X+7)$$

$$Y = \frac{3}{2}X+\frac{7}{2}$$

$$\therefore\ b_{yx} = \frac{3}{2}(>1)$$

The regression equation of X on Y is

$$2X-Y = -1$$

$$2X = Y-1$$

$$X = \frac{1}{2}(Y-1)$$

$$X = \frac{1}{2}Y-\frac{1}{2}$$

$$\therefore\ b_{xy} = \frac{1}{2}$$

The regression coefficients are positive

$$r = \pm\sqrt{b_{xy}.b_{yx}} = \pm\sqrt{\frac{3}{2}\times\frac{1}{2}}$$

$$= \sqrt{\frac{3}{2}\times\frac{1}{2}}$$

$$= \sqrt{\frac{3}{4}}$$

$$= 0.866$$

$$\therefore r = 0.866$$

Example 9.19

For the given lines of regression $3X-2Y=5$ and $X-4Y=7$. Find

(i) Regression coefficients

(ii) Coefficient of correlation

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